China’s New-Energy Vehicle Transition Research

From a sales tipping point to fleet-wide transformation

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Abstract China has passed a major new-energy vehicle (NEV) sales milestone, but the transition of the vehicles already on the road remains much slower. This project combines official market, fleet, charging, powertrain, and export evidence to measure that flow-stock gap. It finds that NEVs reached 50.8% of domestic new-vehicle sales in 2025 but only 12.0% of the registered automobile fleet, a difference of 38.8 percentage points. The project also documents rapid charging-network expansion, a growing PHEV/range-extender share, rising NEV exports, bounded sales-share sensitivities to 2030, and an auditable fleet-turnover model through 2035. Older manually transcribed data were independently checked, source evidence was archived with SHA-256 checksums, and machine-readable schemas plus automated tests enforce data quality. The report emphasizes definitions, uncertainty, bottlenecks, and reproducibility rather than presenting the scenarios as causal forecasts.

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Course	SUM26001 Final Project
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The argument in one sentence: China has crossed the domestic new-sales tipping point for NEVs, but it has not yet transformed the vehicles already on the road.

1 Problem statement and objectives

1.1 The problem

Headlines often treat a high NEV share of new sales as if the whole automobile fleet has already become electric. That interpretation confuses a **flow** with a **stock**. New-vehicle sales measure activity during a period, while the registered fleet contains vehicles accumulated over many years. Charging connectors, powertrain composition, and exports add further denominators that cannot be combined without definition controls.

The central research question is:

How quickly are China's rapidly growing NEV sales translating into changes in the registered vehicle fleet, charging network, powertrain mix, and export market?

1.2 Objectives

The project has six objectives:

1. Build a verified annual automobile and NEV sales series for 2015-2025.
2. Quantify the difference between the NEV share of new sales and the NEV share of the registered fleet.
3. Measure charging-network growth without hiding the 2025 methodology break.
4. Extend the analysis to BEV, PHEV/range-extender, FCEV, and exports.
5. Construct bounded sales-share sensitivities and an auditable fleet-turnover model through 2035.
6. Make the full project reproducible through source records, checksums, data contracts, tests, an executed notebook, Git history, and automated publishing.

2 From the original idea to the final scope

The original proposal focused on a campus survey about student car culture and automotive preferences. That idea was interesting, but it depended on survey recruitment, consent, anonymization, and a response sample that could not be guaranteed within the project period. The project therefore changed from a small primary-data survey to a reproducible public-data study of China's NEV transition.

This was not only a change of topic. It produced a stronger analytical problem: official sources report sales, production, exports, registrations, fleet stock, and charging infrastructure using different scopes. The final project turns those differences into the central research design rather than treating them as inconveniences.

3 Data and evidence

3.1 Source strategy

The project prioritizes official or first-party publications from the China Association of Automobile Manufacturers (CAAM), the Ministry of Industry and Information Technology (MIIT), the National Bureau of Statistics (NBS), the National Energy Administration (NEA), the National Development and Reform Commission (NDRC), and public-security reporting. The annual market series combines historical CAAM charts with later official annual summaries [1], [2], [3], [4], [5], [6], [7], [8], [9], [10].

Fleet observations come from national statistical and public-security sources [11], [12], [13]. Charging observations and the official 2027 target come from the NEA and NDRC [14], [15], [16], [17].

3.2 Dataset inventory

Evidence block	Coverage	Analytical role
Automobile and NEV sales	2015-2025	Scale, growth, and NEV sales share
Registered fleet and charging	2024-2026 snapshots	Flow-stock gap and infrastructure ratios
BEV, PHEV, and FCEV composition	2020-2024	Powertrain mix and reconciliation
Automobile and NEV exports	2021-2025	Export scale and domestic residual cross-check
Current market pulse	H1 and June 2026	Context kept separate from annual observations
Scenario and turnover tables	2025-2035	Bounded sensitivities and stock-flow accounting
Source register and snapshots	18 archived records	Provenance, definitions, and checksum verification

3.3 Data-quality controls

Older manually transcribed values were independently double-entered against separate official evidence. Each archived source capture has a SHA-256 checksum and byte count. A machine-readable data dictionary defines fields, types, ranges, keys, and cross-file rules. Automated tests reject duplicate keys, impossible ranges, unreconciled powertrain totals, impossible export values, invalid probabilities, and stock-flow accounting errors.

These controls matter because the project is small enough for every important number to be inspectable. The goal is not a large opaque database; it is a defensible chain from source to claim.

4 Methods

4.1 Descriptive indicators

The core calculations are deliberately simple:

$$\text{NEV share}_t = \frac{\text{NEV sales}_t}{\text{total automobile sales}_t} \times 100$$

$$\text{YoY growth}_t = \left(\frac{x_t}{x_{t-1}} - 1 \right) \times 100$$

$$\text{CAGR} = \left(\frac{x_{end}}{x_{start}} \right)^{1/n} - 1$$

The 2025 contribution of NEVs to net market growth is the change in NEV sales divided by the change in total automobile sales. A value above 100% is possible when the non-NEV component contracts.

4.2 Bounded sales-share sensitivities

A straight-line share projection can exceed 100%, so the scenario model works in log-odds space:

$$\text{logit } (p_{t+h}) = \text{logit } (p_t) + hg$$

The slower, baseline, and faster sensitivities use gains of 0.15, 0.25, and 0.35. They are illustrations of different adoption paths, not probabilities or causal forecasts. A trend fitted only to 2015-2022 was tested on 2023-2025 to measure form error before presenting future paths.

4.3 Fleet-turnover accounting

The fleet model starts from the observed end-2025 stock and applies this annual identity:

$$\text{ending stock} = \text{opening stock} + \text{inflow} - \text{retirements}$$

The three scenario bundles vary the annual total-fleet turnover rate and the NEV share of domestic inflow. NEVs receive a lower near-term retirement weight because their registered stock is younger. All rates and weights are stored in `data/manual/fleet_turnover_assumptions.csv`, and every modeled row is tested for the stock-flow identity.

5 Results

5.1 Market growth changed scale after 2020

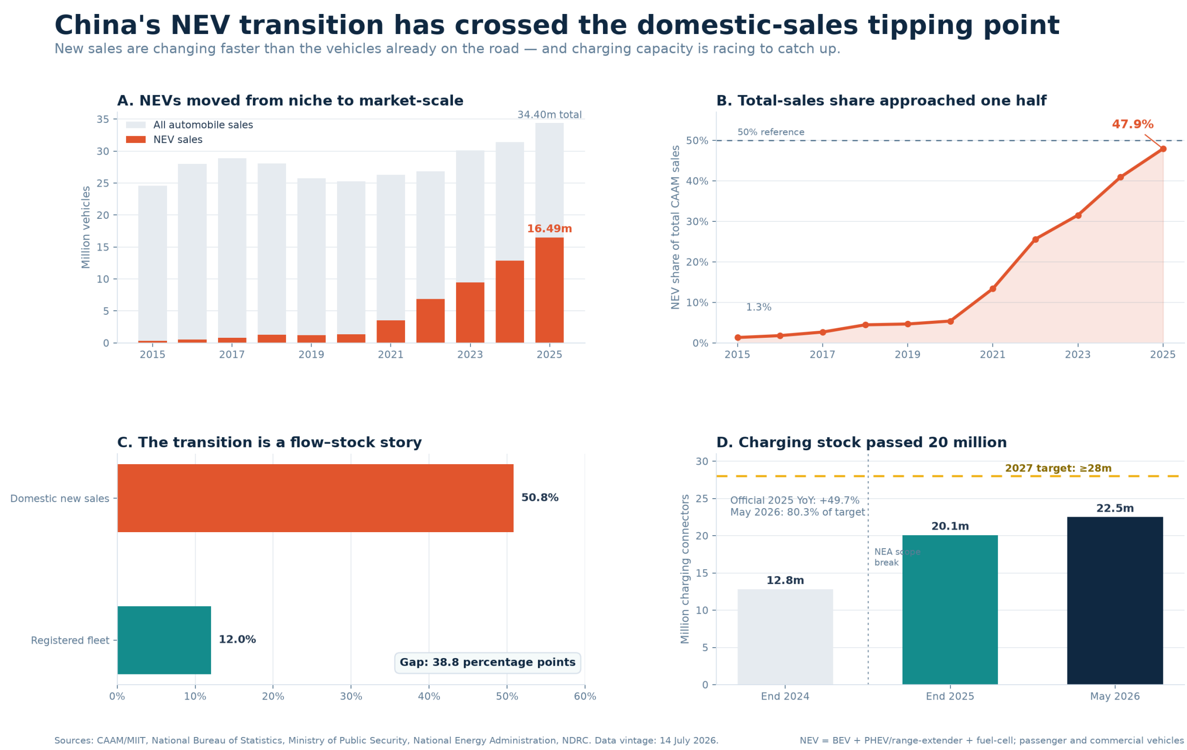

Figure 1: Four views of China's NEV transition: annual sales, total-sales share, the 2025 domestic-sales versus fleet-stock gap, and charging connector levels.

NEV sales rose from **0.331 million in 2015** to **16.490 million in 2025**, equivalent to a **47.8% ten-year CAGR**. The NEV share of total CAAM automobile sales increased from **1.3% to 47.9%**.

Year	Total auto sales (m)	NEV sales (m)	Derived NEV share
2015	24.60	0.33	1.3%
2020	25.31	1.37	5.4%
2021	26.28	3.52	13.4%
2023	30.09	9.50	31.6%
2024	31.44	12.87	40.9%
2025	34.40	16.49	47.9%

From 2024 to 2025, total automobile sales increased by 2.964 million while NEV sales increased by 3.624 million. NEVs therefore accounted for **122.3% of net market growth**; the residual estimate of non-NEV sales fell by 0.660 million.

5.2 The central result is a 38.8-point flow-stock gap

In 2025, NEVs were **50.8% of domestic new-vehicle sales**, but China had only **43.97 million NEVs in a 366.11 million civilian automobile fleet**, or approximately **12.0%** [12], [13]. The difference is **38.8 percentage points**.

This is not a data inconsistency. New sales can change quickly, while the fleet turns over slowly because it contains vehicles registered over many years. The gap is the transition itself.

5.3 Charging expanded rapidly, with a documented break

The NEA reported **20.092 million connectors** at end-2025: 4.717 million public and 15.375 million private. Private connectors were **76.5%** of the network. The stock implied **2.19 registered NEVs per connector**, or **9.32 per public connector** [15].

The agency also introduced a new national reporting system and changed the statistical scope during 2025 [18]. The project therefore does not divide the old-scope end-2024 figure by the revised-scope end-2025 figure. It uses the agency's reported 49.7% growth and marks the break visibly.

By end-May 2026, China had **22.497 million connectors**, already **80.3%** of the official end-2027 target of at least 28 million [16], [17].

5.4 PHEVs changed the mix and exports changed the market boundary

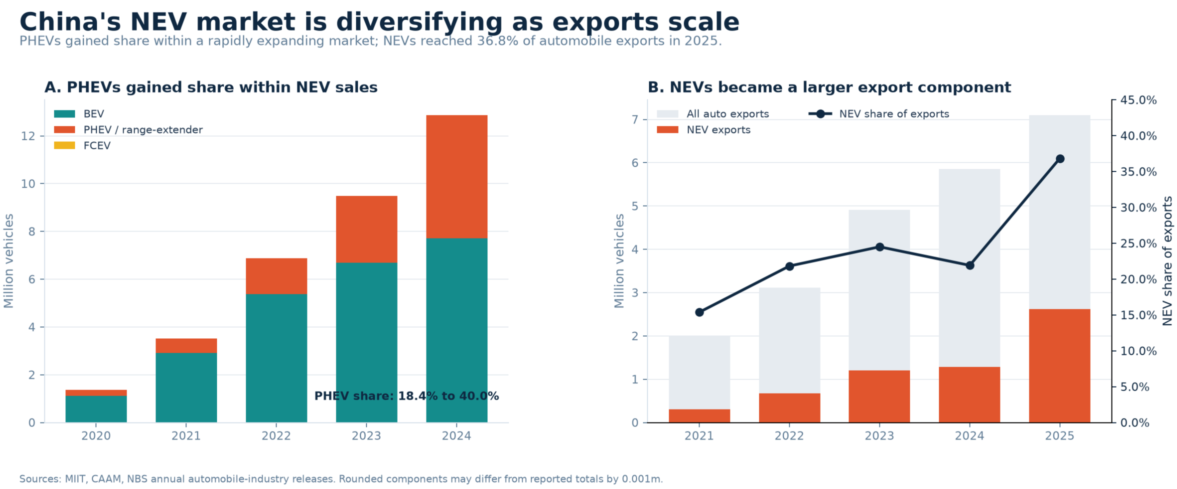

Figure 2: BEV, PHEV/range-extender, and FCEV deliveries alongside total automobile and NEV exports.

Reported BEV deliveries rose from **1.115 million in 2020** to **7.719 million in 2024**, while PHEV/range-extender deliveries rose from **0.251 million** to **5.141 million**. The PHEV share of reported NEV deliveries therefore increased from **18.4% to 40.0%** [4], [6], [8], [19], [20].

Total automobile exports increased from **2.015 million in 2021** to **7.098 million in 2025**, while NEV exports rose from **0.310 million to 2.615 million**. NEVs increased from **15.4% to 36.8% of automobile exports**, and total exports reached **20.6% of CAAM industry deliveries** [21], [22], [23], [24], [25].

Subtracting enterprise-reported exports from industry deliveries produces a 2025 non-export residual NEV share of 50.8%, close to the separately reported domestic share. This is a useful cross-check, not a replacement for official domestic statistics.

5.5 The 2026 pulse is context, not a completed year

CAAM reported **7.446 million NEV sales in the first half of 2026**. The H1 share of total CAAM sales was 49.6%, while the single-month June share reached 58.5% [26], [27]. These observations are displayed separately and never appended to the 2015-2025 annual series.

5.6 Bounded sales-share paths still have form error

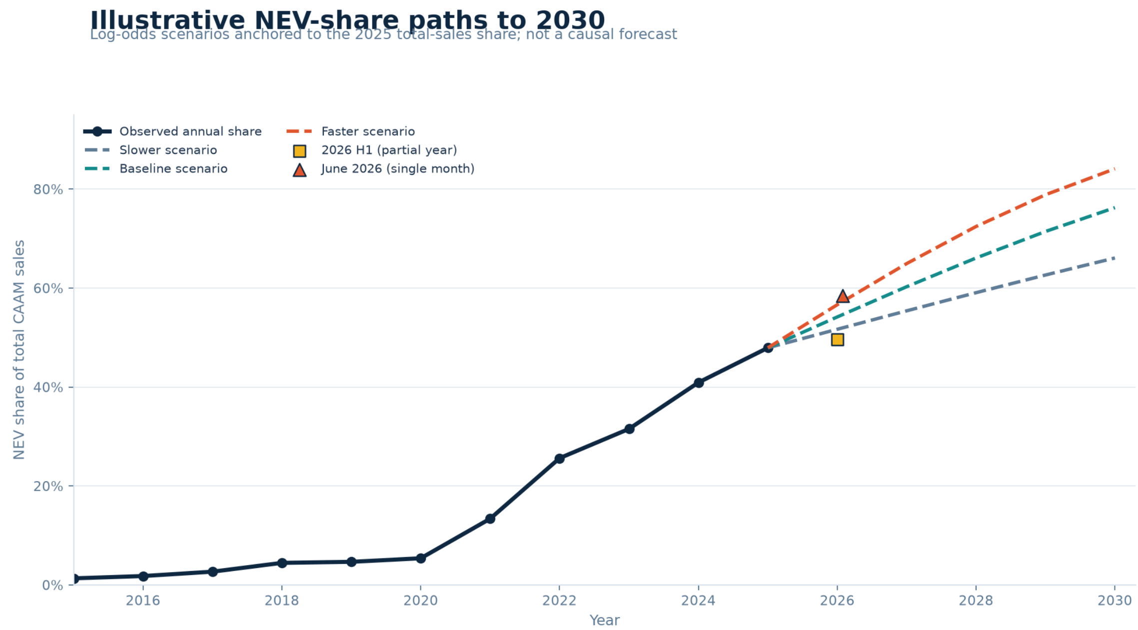

Figure 3: Observed annual NEV sales share followed by three bounded log-odds sensitivities to 2030.

The three sensitivities imply 2030 total-sales shares of approximately **67%, 76%, and 84%**. A held-out test fitted on 2015-2022 and evaluated on 2023-2025 underpredicted all three observations. Its mean absolute error was **5.34 percentage points**. This result is reported to show model limitation, not to claim predictive validation.

5.7 Fleet transformation remains slower than sales transformation

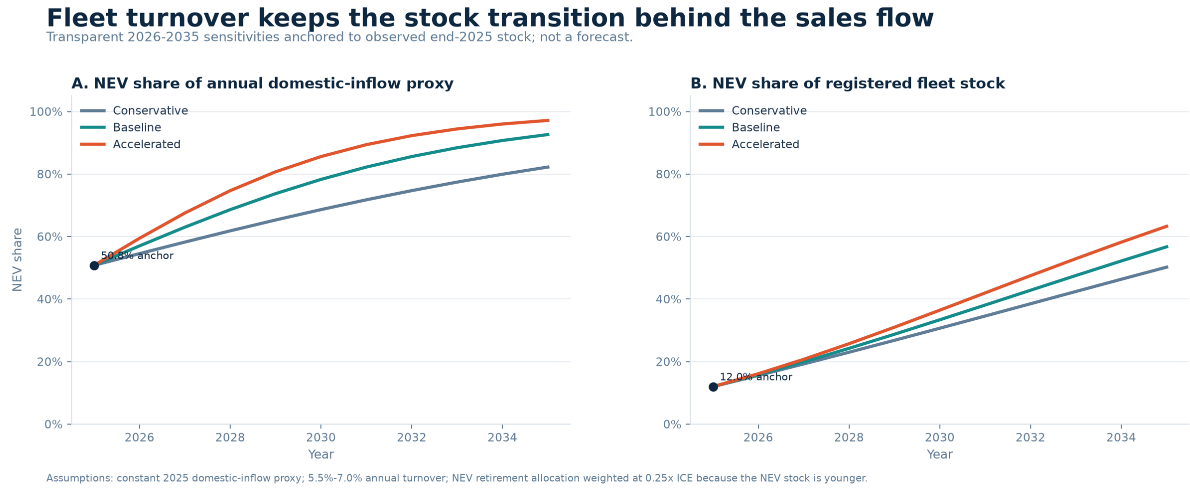

Figure 4: Fleet-turnover sensitivity results for annual domestic inflow share and registered fleet share through 2035.

The baseline NEV share of annual domestic inflow reaches **78.3% in 2030**, but the NEV share of the registered fleet reaches only **33.4%**. By 2035, the baseline values are **92.6% of inflow** and **56.7% of fleet stock**. Across all three bundles, the modeled 2035 fleet share ranges from **50.2% to 63.3%**.

Scenario	2030 NEV inflow share	2030 NEV fleet share	2035 NEV fleet share
Conservative	68.6%	31.6%	50.2%
Baseline	78.3%	33.4%	56.7%
Accelerated	85.6%	35.7%	63.3%

These values quantify the flow-stock lag under explicit assumptions. They are not registration forecasts.

6 Bottlenecks and accommodations

6.1 Bottleneck 1: The original survey could not guarantee sufficient evidence

Core challenge. Measure automotive preferences through a campus survey.

Where the project got stuck. A credible survey required recruitment, consent, anonymization, and enough responses for subgroup comparisons. Those conditions could not be guaranteed within the course schedule.

Change made. The project moved to a national public-data research question with auditable official sources.

Solution found. The new design produced a reproducible longitudinal study and a clearer quantitative contribution: separating new-sales flow from fleet stock.

6.2 Bottleneck 2: Older values were manually transcribed

Core challenge. Use 2015-2020 historical charts without treating one manual entry as unquestioned truth.

Where the project got stuck. Machine-readable official files were not consistently available, and transcription could introduce silent errors.

Change made. Older values were independently double-entered against separate official evidence, with tolerances and pass/fail records.

Solution found. All pre-2021 checks passed, and the audit table now links each comparison to a registered source ID.

6.3 Bottleneck 3: Source pages and definitions can change

Core challenge. Preserve enough evidence for another person to audit the project after publication.

Where the project got stuck. Web pages can be updated, removed, or reported with ambiguous definitions.

Change made. The project added curated text captures, a source register, retrieval dates, definition notes, byte counts, and SHA-256 checksums.

Solution found. A schema validator now verifies every archived capture and connects processed rows to registered source IDs.

6.4 Bottleneck 4: The charging series changed scope

Core challenge. Describe charging growth across 2024-2025.

Where the project got stuck. The NEA changed the statistical scope during 2025, so direct division would create a misleading growth rate.

Change made. The chart displays the break, the report uses the agency's source-reported 49.7% growth, and no calculated cross-break rate is presented.

Solution found. The analysis remains useful while explicitly preserving the limitation. Future work starts a consistent monthly series after the break.

6.5 Bottleneck 5: Windows and Linux normalized text differently

Core challenge. Make archived-source checksums pass both locally and in GitHub Actions.

Where the project got stuck. A Windows working file ended with CRLF while the committed Git blob used LF, causing the first Linux CI run to report a checksum mismatch.

Change made. Snapshot text was explicitly configured as LF, and the manifest was registered against canonical committed-blob bytes.

Solution found. The corrected workflow passed all tests on Linux and then deployed the site. The failed and successful runs remain documented in `PUBLICATION_LOG.md`.

7 Reproducibility and Git evidence

7.1 One-command local rebuild

The reference toolchain uses Python 3.12 and Quarto 1.9.38. Windows users can reproduce analysis, notebook execution, tests, website, final PDF, and validation with:

```
.\scripts\rebuild.ps1
```

The locked cross-platform sequence is:

```
uv sync --frozen --python 3.12
uv run --frozen python scripts/build_analysis.py
uv run --frozen python scripts/build_notebook.py
uv run --frozen python -m nbconvert --to notebook --execute
--inplace notebooks/01_exploration.ipynb --ExecutePreprocessor.timeout=120 --
ExecutePreprocessor.record_timing=False
uv run --frozen python -m unittest discover -s tests -v
quarto render
quarto render final_report.qmd --output final_report.pdf
uv run --frozen python scripts/validate_project.py
```

7.2 Git history as process evidence

The repository was not uploaded as one final file dump. It contains a sequence of focused commits that show how the project developed.

Commit	Stage	Evidence
39de674	Scaffold	Reproducible project structure
b743538	Research design	Question and source protocol
b0618fa	Data	Verified market, fleet, and charging snapshots
b55fe5f	Analysis	Indicators and bounded scenarios
dd2754f	Notebook	Executed exploratory analysis
67851fb	Reporting	Cited Quarto narrative and website
87a8f20	Verification	Independent pre-2021 double-entry audit
e65be68	Provenance	Archived evidence and checksums
f9e0ded	Testing	Machine-readable contracts and schema tests
5587588	Extension	Powertrain and export analysis
ba02911	Modeling	Fleet-turnover model through 2035
0237256	CI repair	Cross-platform checksum policy
358f15f	Publication	v0.2.0 publication evidence
bd2ad4f	Quality repair	Removed corrupted control characters

GitHub Actions independently rebuilds the tables and figures, executes the notebook, runs 20 unit/schema tests, renders both the website and final PDF, validates artifacts, and deploys GitHub Pages.

8 Individual contribution and lessons learned

8.1 Individual contribution

This is a one-member project. **Yang Haoyuan** completed the research design, source collection, data transcription and independent verification, provenance archive, data dictionary, Python analysis, visualizations, powertrain and export extension, bounded scenarios, fleet-turnover model, tests, notebook, Quarto report, Git history, CI workflow, GitHub repository, and Pages publication.

No teammate contribution or merge history is claimed. The commit history is presented as evidence of individual iteration and project management.

8.2 Lessons learned

1. **Definitions are part of the analysis.** Sales, registrations, fleet stock, exports, and charging connectors cannot be compared until their denominators and scopes are explicit.
2. **A high sales share does not equal a transformed fleet.** The flow-stock distinction became the project's strongest explanatory idea.
3. **Reproducibility catches real errors.** Schema tests, checksums, and Linux CI exposed issues that visual inspection alone did not catch.
4. **A limitation should be visible.** Marking the charging methodology break is more credible than manufacturing a smooth but false series.
5. **Models are communication tools when assumptions are visible.** The bounded scenarios and fleet model are useful because their levers, identities, and limits can be inspected; extra decimal places do not make them forecasts.
6. **Project history strengthens the final result.** Small, focused commits made it possible to identify when data, methods, tests, and publication decisions changed.

9 Limitations and future improvements

The project is descriptive and scenario-based. It does not claim that charging construction caused NEV adoption, and it does not treat 2026 partial-year data as a completed annual observation. CAAM industry deliveries, domestic retail sales, registrations, public-security fleet stock, and customs statistics are not interchangeable.

The aggregate turnover model has no vehicle-age cohorts, scrappage-policy shocks, price response, or capacity constraints. Its constant inflow volume is an intentional sensitivity control, not a demand forecast. The next model version should use age-cohort survival functions if suitable registration data become available.

Three practical extensions remain:

1. Build a consistent monthly charging series from the revised reporting system beginning in June 2025.
2. Replace aggregate turnover weights with age-cohort retirement behavior.
3. Add destination-level export analysis only after reconciling customs and enterprise-reporting definitions.

10 Conclusion

China's NEV transition is advanced in new sales but incomplete in the fleet. The 2025 domestic sales share of 50.8% and fleet share of 12.0% describe different stages of the same process. Charging infrastructure, PHEV growth, and exports show that the transition is broader than one sales-share line, while the turnover model explains why vehicles already on the road change more slowly.

The project's main contribution is therefore both empirical and methodological: it quantifies the flow-stock lag and preserves the definition boundaries needed to interpret it. The source archive, machine-readable contracts, tests, Git history, executed notebook, PDF, and public website make that argument auditable rather than merely presentable.

Submission-format checklist

Course requirement	Location or evidence
Problem statement and objectives	Section 1 of this report
Results or bottleneck documentation	Sections 5 and 6
Git collaboration/process evidence	Section 7 and repository history
Storytelling	Original idea, scope change, evidence, results, bottlenecks, lessons, conclusion
Final PDF in repository root	`final_report.pdf`
Generating QMD in repository root	`final_report.qmd`
README project/member/contribution information	`README.md`
Required ignore rules	`.gitignore`

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